



COMPETENCY STANDARD
FOR
MARINE ENGINE ARTIFICER
(SHIPBUILDING SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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The Competency Standards for Marine Engine Artificer is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment. Subsequently, they would be qualified and settled for a relevant job.

Developed under the Skills for Industry Competitiveness and Innovation Program (SICIP), this document addresses the skills requirements critical to marine engine artificer. It is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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INTRODUCTION:

The **Skills for Industry Competitiveness and Innovation Program (SICIP)** has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from Construction industry in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at Glance:**Generic Competencies (20 hrs.)**

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-MEA-01-G	Perform Computations Using Basic Mathematical Formulas	1. Identify calculation requirements in the workplace 2. Identify and select appropriate mathematical methods. 3. Use tool/instrument to perform calculations	10
SICIP-SBD-MEA-02-G	Communicate in English in the Workplace	1. Read and understand workplace documents in English 2. Prepare simple workplace documents in English 3. Listen and comprehend to English conversations 4. Perform conversations in English language	05
SICIP-SBD-MEA-03-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	05
Total Hour			20 Hrs.

Sector Specific Competencies (60 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-SBD-MEA-01-S	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-SBD-MEA-02-S	Translate Manuals, Mechanical Drawings, Specifications and Estimation	1. Understand symbols, mechanical drawing, specifications and estimation 2. Interpret manuals, mechanical drawings specific to shipbuilding works 3. Store manuals, drawings and plans	14
SICIP-SBD-MEA-03-S	Work with Hand Tools, Power Tools	1. Inspect hand tools and power tools for usability 2. Use hand tools properly and safely 3. Operate power tools properly and safely 4. Clean/maintain hand tools and power tools after use	16
SICIP-SBD-MEA-04-S	Carry Out Measurements and Calculations	1. Check usability of measuring devices 2. Carry out accurate work measurements 3. Execute simple work calculations 4. Clean and maintain measuring instruments	16
SICIP-SBD-MEA-05-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	04
Total Hour			60 Hrs.

OCCUPATION SPECIFIC (CORE) COMPETENCIES (280 HRS.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-SBD-MEA-01-O	Demonstrate Knowledge of Marine Diesel Engines and Systems	1 Interpret working principles of two-stroke and four-stroke diesel engines 2 Illustrate major components and system of diesel engine 3 Explain operation of diesel engine	60
SICIP-SBD-MEA-02-O	Perform Combustion System Servicing	1 Carry out piston servicing and maintenance 2 Carry out cylinder head servicing and maintenance 3 Service cylinder liner 4 Perform main and crank pin bearing clearance check	80
SICIP-SBD-MEA-03-O	Perform Engine Oil System Servicing	1 Carry out engine oil replacement 2 Service engine oil pump 3 Service and maintain engine oil cooler	40
SICIP-SBD-MEA-04-O	Perform Cooling System Servicing	1 Carry out cooling chamber maintenance 2 Service cooling pump 3 Service and maintain water cooler	60
SICIP-SBD-MEA-05-O	Perform Maintenance of Intake, Exhaust System	1 Carry out air cooler maintenance 2 Carry out exhaust manifold maintenance	40
Total Hours			280 Hrs.

COMPETENCY STANDARD: MARINE ENGINE ARTIFICER

The Generic Competencies

Unit of Competency: PERFORM COMPUTATIONS USING BASIC MATHEMATICAL FORMULAS	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-MEA-01-G
Unit Descriptor: This unit of competency requires knowledge, skills and attitude to perform computations using basic mathematical formulas. It specifically includes the tasks of identifying calculation requirements in the workplace, identifying and selecting appropriate mathematical methods, and using appropriate tools and instruments to perform calculations.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify calculation requirements in the workplace	1.1 <u>Calculation requirements</u> are identified 1.2 <u>Workplace informations</u> are interpreted
2. Identify and select appropriate mathematical methods	2.1 <u>Appropriate calculation method</u> is identified 2.2 Appropriate calculation method is selected
3. Use tool/instrument and perform calculations	3.1 Appropriate <u>tools and instruments</u> are identified. 3.2 Computation is made

Range of variables:

Variable	Range May include but not limited to:
1. Calculation requirements.	1.1 Area 1.2 Volume 1.3 Measurements: Length/Width/Height/thickness/diameter 1.4 Weight and density Calculation
2. Workplace information	2.1 Drawing 2.2 Part drawing 2.3 Verbal instructions 2.4 Job order and Job sheet
3. Appropriate calculation method	3.1 Addition, Subtraction, Division and Multiplication 3.2 Conversion of measurement 3.3 Percentage and ratio calculation 3.4 Geometric Formulae (Square, Rectangle, Circle, Sphere etc)
4. Tools/instruments	4.1 Calculator 4.2 Computer

Curricular Content Guide

1. Underpinning Knowledge	1.1 Numerical concept, identification of calculation requirements 1.2 Basic mathematical methods such as addition, subtraction, multiplication, division, percentage, conversion, ratio and basic geometry formulas. 1.3 Mathematical symbols. 1.4 Measuring units
2. Underpinning Skills	2.1 Computing basic mathematics. 2.2 Practicing conversion of measurements 2.3 Calculating areas, volume etc. using formulas
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect to peers, sub-ordinates and seniors in workplace 3.5 Environmental concern 3.6 Sincerity and honesty
4. Resource Implications	The following resources must be provided. 4.1 Stationeries 4.2 Consumables 4.3 Calculators 4.4 Computers 4.5 Measuring tape

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified calculation requirements from workplace information. 1.2 Selected appropriate method to carry out the calculation requirements. 1.3 Completed calculations using appropriate tools/instruments.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Oral questioning 2.3 Demonstration.
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: COMMUNICATE IN ENGLISH IN THE WORKPLACE	Nominal Duration: 05 hrs.	Unit Code: SICIP-SBD-MEA-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to communicate in English in the workplace. It specifically includes the tasks of reading and understanding workplace documents in English, preparing simple workplace documents in English, listening and comprehending to English conversations and performing conversations in English.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Read and understand workplace documents in English	1.1 Workplace documents are read and understood 1.2 Visual information is interpreted.
2. Prepare simple workplace documents in English	2.1 Simple <u>routine workplace documents</u> are prepared using key words, phrases, simple sentences 2.2 <u>Visual aids</u> are prepared and submitted. 2.3 Key information is written in the appropriate places in standard forms.
3. Listen and comprehend to English conversations	3.1 Active listening is demonstrated. 3.2 English conversation is comprehended and shared
4. Perform conversations in English language	4.1 Conversation is performed in English with peers, customers and management as per the workplace standard.

Range of Variables

Variable	Range May Include but not limited to:
1. Routine workplace documents	1.1 Agenda 1.2 CV/Resume 1.3 Simple reports such as progress and incident reports 1.4 Job sheets 1.5 Standard operating procedure (SOP) 1.6 Operational manuals 1.7 Brochures and promotional material 1.8 Visual and graphic materials 1.9 OSH information 1.10 Signs
2. Visual aids	2.1 Maps 2.2 Diagrams 2.3 Forms 2.4 Labels 2.5 Graphs

	2.6 Charts
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Curricular Content Guide:

1. Underpinning Knowledge	1.1 Workplace documents in English 1.2 Simple routine workplace documents 1.3 Standard operating procedure 1.4 Job roles, responsibilities and compliances
2. Underpinning Skills	2.1 Ability to read and understand workplace documents in English by using appropriate vocabulary and grammar, standard spelling and punctuation 2.2 Ability to write simple routine workplace documents in English such as: Schedules and agenda, job sheets, operational manuals and brochures and promotional material 2.3 Ability of listening in English and interpreting 2.4 Ability to perform conversation in English with peers, customers and management 2.5 Work effectively with others 2.5.1 Listening and questioning skills 2.5.2 Ability to follow simple directions
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Promptness in carrying out activities 3.3 Tidiness and timeliness 3.4 Respect of peers, sub-ordinates and seniors in workplace 3.5 Environmental concern 3.6 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Work place Procedure 4.2 Materials relevant to the proposed activity 4.3 All tools, equipment, material and documentation required. 4.4 Relevant specifications or work instructions

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Conversed in English with peers and customers. 1.2 Made reports of workplace documents in English.
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral questioning 2.4 Interview

3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: OPERATE IN A SELF-DIRECTED TEAM	Nominal Duration: 05 hrs.	Unit Code: SICIP-SBD-MEA-03-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are demonstrated.
2. Communicate and cooperate with team members.	2.1 <u>Effective interpersonal skills</u> are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal forms of communication are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member.	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practice. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range May Include but not limited to:
1. Effective interpersonal skills	1.1 Communication Skills 1.2 Teamwork & Collaboration 1.3 Problem-Solving & Critical Thinking 1.4 Adaptability & Flexibility 1.5 Conflict Resolution & Negotiation 1.6 Emotional Intelligence (EQ) 1.7 Leadership & Mentorship 1.8 Time Management & Organization 1.9 Professionalism & Work Ethic

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Effective interpersonal skills to interact with team members 1.4 Formal and informal forms of communication 1.5 Workplace terminology 1.6 Problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Demonstrating roles and responsibilities of team members 2.3 Interacting with team members and to contribute to activities 2.4 Practicing formal and informal communication 2.5 Performing tasks as per team requirements and workplace practice. 2.6 Solving problems and evaluating outcome of the solution
3. Underpinning Attitudes	2.1 Teamwork 2.2 Promptness in carrying out activities. 2.3 Tidiness and timeliness. 2.4 Respect of peers, sub-ordinates and seniors in workplace. 2.5 Sincere and honest to duties
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Pens 4.3 Papers 4.4 Learning manuals

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified team goals and work processes 1.2 Communicated and cooperated with team members. 1.3 Worked as a team member 1.4 Solved problems as a team member
2. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral questioning
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

The Sector Specific Competencies

Unit of Competency: APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	Nominal Duration: 10 hrs.	Unit Code: SICIP-SBD-MEA-01-S
Unit Descriptor: This unit covers knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks and responding to emergencies.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 <u>OHS policies</u> and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 <u>Personal Protective Equipment (PPE)</u> is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 <u>Hazards and risks</u> are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 <u>Emergency response plans and procedures</u> are implemented. 4.3 <u>First aid procedure</u> is applied during emergency situations.

Range of Variables

Variable	Range
	May include but not limited to:
1. OHS policies	1.1 International OHS policies 1.2 Local OHS requirements 1.3 Fire Safety Rules and Regulations

	1.4 Industry safety Guidelines
2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Safety glasses 2.7 Ear plugs 2.8 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards. 3.2 Biological hazards. 3.3 Physical Hazards. 3.3.1 Machine hazards. 3.3.2 Materials hazards. 3.3.3 Tools and Equipment hazards.
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures. 1.2 Work safety procedures. 1.3 Emergency procedures. 1.3.1 Firefighting. 1.3.2 Earthquake response. 1.3.3 Explosion response. 1.3.4 Accident response. 1.4 Types of (biological, chemical and physical) and their effects. 1.5 PPE types and uses. 1.6 Personal hygiene practices. 1.7 OHS awareness.
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Following personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices

	3.2 Communication with peers, sub-ordinates and seniors in workplace 3.3 Promptness in carrying out activities 3.4 Tidiness and timeliness 3.5 Respect of peers, sub-ordinates and seniors in workplace 3.6 Environmental concern 3.7 Sincere and honest to duties
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 PPEs 4.3 Firefighting equipment 4.4 Emergency response manual 4.5 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Followed OHS policies and procedures. 1.2 Selected and used personal protective equipment (PPE). 1.3 Practiced personal health and safety procedures 1.4 Reported incidents arising from hazards and risks to authority. 1.5 Implemented plans and procedures to respond emergency 1.6 Applied basic first aid procedure.
3. Methods of Assessment	Methods of assessment may include but not limited to: 2.1 Written test 2.2 Demonstration 2.3 Oral questioning 2.4 Interview
4. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: TRANSLATE MANUALS, MECHANICAL DRAWINGS, SPECIFICATIONS AND ESTIMATION	Nominal Duration: 14 hrs.	Unit Code: SICIP-SBD-MEA-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to translate manuals, schematic drawings, specifications and work estimation. It specifically includes the tasks of understanding symbols, mechanical drawing, specifications, and work estimation, interpreting manuals, drawings specific to Engineering works and storing manuals, drawings and plans.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand symbols, mechanical drawing, specifications and estimation	1.1 Symbols are identified and interpreted 1.2 Drawing and specifications are identified 1.3 Estimation according to drawing and specifications is communicated
2. Interpret manuals, mechanical drawings specific to shipbuilding works	2.1 <u>Manuals</u> and mechanical drawings are recognized from design and plan. 2.2 Terms and abbreviations in drawings are recognized. 2.3 Signs and symbols associated with the mechanical components and devices in the drawing are interpreted.
3. Store manuals, drawings and plans	3.1 Manuals, drawings, and plans are collected. 3.2 Manuals, drawings and plans are stored to prevent damage, and ready access and updating of information when required.

Range of Variables

Variable	Range
	May include but not limited to:
1. Manuals	1.1 Manufacturer's Specification Manual 1.2 Repair Manual 1.3 Maintenance Procedure Manual 1.4 Quality Manual 1.5 Instruction Manual

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Types of Manuals 1.2 Estimation of mechanical components/device and other associated materials. 1.3 Signs and Symbols associated with the mechanical components and devices
2. Underpinning Skills	2.1 Checking appropriate version of the manual

	2.2 Identifying relevant drawings and specifications 2.3 Identifying signs and symbols 2.4 Storing manuals
3. Underpinning Attitudes	3.1 Eagerness to learn 3.2 Orderliness 3.3 Resourcefulness
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Shipbuilding manuals and literatures 4.3 Pens 4.4 Papers 4.5 Work books

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Checked version and date of the manual 1.2 Identified relevant drawings and specifications correctly. 1.3 Identified terms and abbreviations. 1.4 identified Signs and symbols. 1.5 Interpreted drawings and specifications.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation 2.5 Portfolio
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: WORK WITH HAND TOOLS AND POWER TOOLS	Nominal Duration: 16 hrs.	Unit Code: SICIP-SBD-MEA-03-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work with hand tools and power tools properly. It specifically includes the tasks of inspecting hand tools and power tools for usability, using hand tools properly and safely, operating power tools properly and safely and cleaning/maintaining hand tools and power tools after use.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Inspect hand tools and power tools for usability	1.1 Appropriate tools are selected. 1.2 Application of tools to job requirement is determined. 1.3 Usability of tools are checked and verified. 1.4 <u>Hand tools</u> and <u>power tools</u> are prepared. 1.5 Sources of power supply for power tools identified.
2. Use hand tools properly and safely	2.1 Appropriate hand tool for the job is used. 2.2 Proper and safe use/operation is applied in the different types of hand tools. 2.3 <u>Safety precautions</u> is observed when using hand tools. 2.4 Unsafe or faulty tools are identified and marked for repairing
3. Operate power tools properly and safely	3.1 Power supply outlet and electrical cord are inspected and confirmed safe for use in accordance with established workplace safety requirements. 3.2 Proper sequence of operation is applied in using power tools to produce results. 3.3 Power tools are used safely in accordance to manufacturer's operating specification.
4. Clean/maintain hand tools and power tools after use	4.1 Dust and foreign materials are removed from power tools in accordance with workplace standard. 4.2 Scraps or wastages are removed and cleaned 4.3 Condition of tools is checked after use. 4.4 Appropriate lubricant is applied after use and prior to storage. 4.5 <u>Measuring tools</u> are checked and calibrated. 4.6 Defective tools, instruments, power tools and accessories are inspected and corrected or replaced.

Range of Variables

Variable	Range
1. Hand tools	May include but not limited to: 1.1 Screwdrivers

	1.2 Pliers 1.3 Wire Strippers 1.4 Fish Wire (for pulling wire through conduit) 1.5 Leveler 1.6 Tester 1.7 Allen key 1.8 Nut Drivers 1.9 Crimping Tool (for wire terminals) 1.10 Flashlight or Headlamp 1.11 Hacksaw (for cutting conduit) 1.12 Chisels and Hammer 1.13 Adjustable wrench 1.14 File 1.15 Socket wrench 1.16 Ratchet 1.17 Torque wrench 1.18 Pipe wrench 1.19 Circlip plier 1.20 Chain block 1.21 Piston ring opener
2. Power tools	2.1 Power drill 2.2 Nail gun 2.3 Angle grinder 2.4 Pneumatic wrench 2.5 Pedestal drill 2.6 Power brushing tool 2.7 Hydraulic jack
3. Safety precautions	3.1 Use of appropriate PPEs 3.2 Proper hand, feet and eye coordination 3.3 Working environment 3.4 Safe operating condition of hand tools and power tools 3.5 Awareness to OHS requirements
4. Measuring instruments	4.1 Tachometer 4.2 Vernier Calipers 4.3 Calipers 4.4 Inside Micrometer 4.5 Outside micrometer 4.6 Bevel Protractor 4.7 Deflection gauge 4.8 Feeler gauge 4.9 Telescopic feeler gauge 4.10 Multimeter 4.11 Clamp meter

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types of Hand tools and their proper use and techniques 1.2 Types of Power tools, use and safe handling method
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	1.3 Procedures in the use of hand tools and power tool 1.4 Reporting and documentation 1.5 Preventive maintenance 1.6 Quality procedures 1.7 Waste management 1.8 Storage procedures
2. Underpinning Skills	2.1 Using appropriate hand tool for the job 2.2 Observing safety precautions when using hand tools 2.3 Using power tools correctly and safely in accordance to manufacturer's operating specification. 2.4 Checking condition of tools after use 2.5 Applying appropriate lubricant on hand tools and power tools after use and prior to storage 2.6 Inspecting and correcting or replacing defective tools, instruments, power tools and accessories 2.7 Storing Tools and power tools safely in appropriate location
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety practices 3.2 Environmental concerns 3.3 Eagerness to learn 3.4 Tidiness and timeliness 3.5 Concern to proper use of tools 3.6 Orderliness
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Shipbuilding hand tools and power tools 4.3 Pens 4.4 Papers 4.5 Work books 4.6 Operation and maintenance manuals

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Used appropriate hand tools for the job. 1.2 Observed safety precautions when using hand tools. 1.3 Used power tools safely in accordance to manufacturer's operating specification. 1.4 Cleaned and maintained hand tools and power tools after use and prior to storage. 1.5 Inspected and corrected or replaced defective tools, instruments, power tools and accessories. 1.6 Stored tools and power tools safely in appropriate location.
2. Methods of Assessment	Competency should be assessed by:

	2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: CARRY OUT MEASUREMENTS AND CALCULATIONS	Nominal Duration: 16 hrs.	Unit Code: SICIP-SBD-MEA-04-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to carry-out measurements and calculations. It specifically includes the tasks of checking usability of measuring devices, carrying out accurate work measurements, executing simple work calculations and cleaning and maintaining measuring instruments.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Check usability of measuring devices	1.1 Appropriate <u>measuring device</u> is selected for the job. 1.2 Applications of measuring device is determined. 1.3 Usability of measuring device is checked and verified. 1.4 Measuring device is prepared.
2. Carry out accurate work measurements	2.1 Measurements are obtained using appropriate measuring device. 2.2 <u>Systems of measurements</u> are identified and converted where necessary. 2.3 Results are confirmed and recorded.
3. Execute simple work calculations	3.1 Simple calculations involving <u>four basic mathematical calculations</u> are executed. 3.2 Other operations are used to complete tasks in Shipbuilding works 3.3 Appropriate formulas for calculating quantities of materials are selected. 3.4 Calculations are performed and verified. 3.5 Material quantities are calculated if required. 3.6 Results are interpreted and communicated to authority.
4. Clean and maintain measuring instruments	4.1 Dust and foreign materials are removed from measuring instrument 4.2 Condition of instrument is checked 4.3 Appropriate lubricant is applied after use and prior to storage 4.4 Measuring instruments are checked and calibrated. 4.5 Instrument is stored in accordance to workplace procedure

Range of Variables

Variable	Range
	May include but not limited to:
1. Measuring device	1.1 Slide calipers 1.2 Steel tape measure 1.3 Steel rule

	1.4 Square 1.5 Tri-square 1.6 Feeler gauges 1.7 Water level 1.8 Hose level 1.9 Thermometers 1.10 Protractor
2. Systems of measurements	2.1 Metric system 2.2 FPS system
3. Four basic mathematical Calculation	3.1 Area Calculation 3.2 Volume Calculation 3.3 Weight Calculation 3.4 Angle Calculation

Curricular Content Guide

1. Underpinning Knowledge	1.1 Types and principles of operation of measuring devices 1.2 The ISO standard of measurements 1.3 Methods of measurement and calculation 1.4 Fraction and decimals 1.5 Linear measurement 1.6 Units of conversion and conversion factors in measurements 1.7 Dimensioning and fits and tolerances 1.8 Calculating ratio and proportion 1.9 Care in the use of measuring devices
2. Underpinning Skills	2.1 Selecting appropriate measuring device for the job 2.2 Checking and verifying usability of measuring device 2.3 Obtaining measurements using appropriate measuring device. 2.4 Confirming measurements and recording results 2.5 Carrying out simple calculations involving four basic mathematical operations 2.6 Calculating material quantities 2.7 Interpreting and communicating results to authority 2.8 Cleaning and storing measuring instruments
3. Underpinning Attitudes	3.1 Cleanliness/tidiness 3.2 Commitment to occupational health and safety practices 3.3 Environmental concerns 3.4 Eagerness to learn 3.5 Timeliness and orderliness 3.6 Respect for rights of peers and seniors in workplace
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of measuring and checking tools/instruments 4.3 Pens 4.4 Papers

	4.5 Work books
	4.6 Measuring tools' operating and maintenance manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Selected appropriate measuring device for the job. 1.2 Checked and verified usability of measuring device. 1.3 Obtained measurements using appropriate measuring device. 1.4 Confirmed measurements and recorded results. 1.5 Carried out Simple calculations involving four basic mathematical operations. 1.6 Calculated material quantities. 1.7 Interpreted and communicated Results to authority.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: APPLY GREEN PRACTICES	Nominal Duration: 04 hrs.	Unit Code: SICIP-SBD-MEA-05-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace, and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 <u>Principles of green practices</u> are explained. 1.2 <u>Sources of environmental impacts</u> during shipbuilding activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 <u>Ways of mitigating environmental impacts</u> in shipbuilding sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 <u>Recyclable and non-recyclable items</u> are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 <u>Different types of waste</u> are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range
	May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 use of sustainable materials with low environmental impact

	1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in Shipbuilding sector
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	1.2 Key terms and symbols related to environmental sustainability in Ship drawings 1.3 Sources of environmental impacts in Shipbuilding 1.4 Methods to minimize resource consumption (water, energy, raw materials) 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in Shipbuilding activities 2.2 Applying methods to minimize resource use in the workplace 2.3 Implementing waste management practices 2.4 Following procedures to safely dispose of hazardous materials 2.5 Documenting water, energy, and material consumption in the workplace 2.6 Performing green practices in personal and professional activities
3. Underpinning Attitudes	3.1 Commitment to sustainable and eco-friendly Shipbuilding practices 3.2 Active participation in reducing resource use and waste generation 3.3 Eagerness to adopt green habits in professional life 3.4 Promptness in implementing sustainable practices in Shipbuilding activities 3.5 Tidiness and concern for environmental cleanliness 3.6 Compliance with safety standards and environmental regulations
4. Resource Implications	4.1 Workplace (simulated or actual) 4.2 Different types of Shipbuilding hand tools and power tools 4.3 Safety gear 4.4 Pens 4.5 Papers 4.6 Work books 4.7 Operation and maintenance manuals 4.8 Waste segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Explained the principles of green practices in Shipbuilding 1.2 Identified sources of environmental impacts
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	1.3 Implemented procedures to minimize resource use (water, energy, raw materials) 1.4 Disposed of hazardous waste in compliance with safety and environmental standards 1.5 Practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-specific Competencies

Unit of Competency: DEMONSTRATE KNOWLEDGE OF MARINE DIESEL ENGINES AND SYSTEMS	Nominal Duration: 60 hrs.	Unit Code: SICIP-SBD-MEA-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to demonstrate knowledge of marine diesel engines and systems. It specifically includes the tasks of Interpreting working principles of two-stroke and four-stroke diesel engines, illustrating major components and system of diesel engine and explaining operation of diesel engine.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret working principles of two-stroke and four-stroke diesel engines	1.1 The functionality of two-stroke and four-stroke engines are described 1.2 Two-stroke and four-stroke cycle diesel engines are compared 1.3 Pressure-charging principles are explained 1.4 <u>Engine combustion characteristics</u> are illustrated 1.5 <u>Common combustion faults</u> are explained
2. Illustrate major components and system of diesel engine	2.1. <u>Major components of diesel engine</u> are identified 2.2. Functions of major components of diesel engine are explained 2.3. <u>Major engine systems</u> are identified and explained
3. Explain operation of diesel engine	3.1. Starting system of diesel engine is explained 3.2. <u>Operational parameters</u> of diesel engines are explained 3.3. Emergency isolation and stopping procedures are clarified

Range of Variables

Variable	Range
	May include but not limited to:
1. Engine combustion characteristics	1.1 Cylinder Pressure 1.2 Heat Release Rate 1.3 Ignition Delay and diesel knock 1.4 Combustion Duration 1.5 Maximum Pressure Rise Rate
2. Common combustion faults	2.1 Insufficient air and fuel supply 2.2 Incomplete combustion

	2.3 Exhaust temperature rise
3. Major components of diesel engine	3.1 Cylinder Block, liner & Head 3.2 Piston Assembly (crown, connecting rod, ring) 3.3 Crankshaft and bearings 3.4 Camshaft
4. Major engine systems	4.1 Fuel System 4.2 Cooling System 4.3 Lubrication System 4.4 Electrical System 4.5 Air Intake & Exhaust System 4.6 Scavenging system
5. Operational parameters	5.1 Fuel temperature and pressure 5.2 Engine Oil temperature and pressure 5.3 Exhaust gas temperature 5.4 Scavenge air temperature 5.5 Cooling water temperature and pressure 5.6 Battery (Specific gravity of electrolyte, terminal voltage)

Curricular Content Guide

1. Underpinning knowledge	1.1 Functions and difference of two-stroke and four-stroke diesel engines 1.2 Principles of pressure charging 1.3 Characteristics of engine combustion 1.4 Major components and systems of diesel engine 1.5 Operational parameters of diesel engines
2. Underpinning Skills	2.1 Identifying Major components of diesel engine 2.2 Identifying Common combustion faults 2.3 Illustrating engine system 2.4 Demonstrating operational parameter
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness
4 Resource Implications	4.1. Marine operations site 4.2. Marine diesel engines and systems 4.3. Tools, equipment and personal protective equipment 4.4. Appropriate workplace. 4.5. CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified Major components of diesel engine 1.2 Identified Common combustion faults 1.3 Illustrated engine system 1.4 Demonstrated operational parameter
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM COMBUSTION SYSTEM SERVICING	Nominal Duration: 80 hrs.	Unit Code: SICIP-SBD-MEA-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform combustion system servicing. It specifically includes the tasks of carrying out piston servicing and maintenance, carrying out cylinder head servicing and maintenance, servicing cylinder liner, performing main and crank pin bearing clearance check.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out piston servicing and maintenance	1.1 <u>Piston</u> of the engine is opened and disassembled 1.2 <u>Piston rings</u> are checked and serviced or replaced 1.3 Piston crown is cleaned as per job requirement. 1.4 Piston pin is opened and its clearance is checked and replaced 1.5 Piston is assembled and fitted back as per instruction manual
2. Carry out cylinder head servicing and maintenance	2.1 <u>Exhaust and inlet valves</u> are opened, cleaned, checked 2.2 Valves are lapped or replaced and fitted back to the cylinder head 2.3 Cooling space are cleaned
3. Service cylinder liner	3.1. <u>Cylinder liner</u> wear down is measured and checked 3.2. Cylinder liner is replaced if required 3.3. Mating face between cylinder liner and head is checked and lapped if required.
4. Perform main and crank pin bearing clearance check	4.1 Main and crank pin bearing are identified 4.2 Main and crank pin bearing clearance are checked as per instruction manual 4.3 Reasons for bearing wear down are investigated and rectified
5. Clean and store tools and equipment.	5.1 Workplace, tools and equipment are cleaned and maintained in accordance with workplace requirements 5.2 Waste materials are disposed in proper place. 5.3 Tools, equipment and finished products are stored safely in accordance with workplace procedures

Range of Variables

Variable	Range
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	May include but not limited to:
1 Piston	1.1 Crosshead piston for two stroke engines 1.2 Trunk type piston for four stroke engines
2 Piston rings	2.1 Compression ring 2.2 Oil scraper ring
3 Exhaust and inlet valves	3.1 Stainless steel valve seat 3.2 Nimonic steel valve seat 3.3 Rocker arm and push rod
4 Cylinder liner	4.1 Uniflow cylinder liner 4.2 Cross flow liner

Curricular Content Guide

1 Underpinning knowledge	1.1 Engine Components and Functions 1.2 Disassembly and Assembly Procedures of piston 1.3 Engine Valve System and Functions 1.4 Valve Lapping and Replacement 1.5 Cylinder Liner Function and Importance 1.6 Cylinder Liner Replacement Procedures 1.7 Mating Face Inspection and Lapping 1.8 Main and Crank Pin Bearings Function and Importance 1.9 Bearing Identification and Types 1.10 Investigation and Troubleshooting of Bearing Wear
2 Underpinning Skills	2.1 Performing disassembly and assembly the piston 2.2 Performing disassembly and assembly exhaust and inlet valves 2.3 Checking cylinder liner wear using appropriate tools 2.4 Inspecting and lapping the mating face between the cylinder liner and head 2.5 Identifying main and crank pin bearings in the engine 2.6 Checking bearing clearance using appropriate tools 2.7 Investigating causes of bearing wear
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness
4 Resource Implications	4.1. Marine operations site 4.2. Marine diesel engines and systems 4.3. Tools, equipment and personal protective equipment 4.4. Appropriate workplace.

	4.5. CBLM
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Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Performed disassembly and assembly the piston 1.2 Performed disassembly and assembly exhaust and inlet valves 1.3 Checked cylinder liner wear using appropriate tools 1.4 Inspected and lapping the mating face between the cylinder liner and head 1.5 Identified main and crank pin bearings in the engine 1.6 Checked bearing clearance using appropriate tools
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM ENGINE OIL SYSTEM SERVICING	Nominal Duration: 40 hrs.	Unit Code: SICIP-SBD-MEA-03-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform engine oil system servicing. It specifically includes the tasks of carrying out engine oil replacement, servicing engine oil pump and Servicing and maintaining engine oil cooler.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out engine oil replacement	1.1 Existing <u>engine oil</u> is drained and sump is cleaned 1.2 Engine oil filter is replaced and fresh oil is poured/renewed
2. Service engine oil pump	2.1 Engine oil main pump and priming pump are inspected 2.2 Engine oil pumps are opened and disassembled 2.3 Servicing of Engine oil pumps are performed and fitted back
3. Service and maintain engine oil cooler	3.1 <u>Engine oil cooler</u> is checked and opened 3.2 Engine oil cooler serviced and cleaned

Range of Variables

Variable	Range May include but not limited to:
1 Engine oil	1.1 Mineral-Based Two-Stroke Oil 1.2 Synthetic Two-Stroke Oil 1.3 Biodegradable Two-Stroke Oil 1.4 Mineral-Based Four-Stroke Oil 1.5 Synthetic Blend (Semi-Synthetic) Four-Stroke Oil 1.6 Fully Synthetic Four-Stroke Oil 1.7 High-Performance Diesel Marine Oil
2 Engine oil cooler	2.1 Shell and Tube Oil Cooler 2.2 Plate-Type Oil Cooler

Curricular Content Guide

1 Underpinning knowledge	1.1 Types and properties of engine oils (mineral, synthetic, biodegradable, etc.). 1.2 Functions and components of engine oil systems, including pumps and coolers.
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	1.3 Proper procedures for oil replacement and system servicing. 1.4 Causes and effects of oil contamination and degradation. 1.5 Workplace safety procedures and environmental regulations for oil disposal.
2 Underpinning Skills	2.1 Performing Draining and replacing engine oil efficiently. 2.2 Cleaning and servicing engine oil filters and sumps. 2.3 Performing Disassembling, inspecting, and servicing engine oil pumps. 2.4 Checking, cleaning, and maintaining engine oil coolers. 2.5 Using appropriate tools and following servicing procedures accurately.
3 Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness
4 Resource Implications	4.1 Marine operations site 4.2 Marine diesel engines and systems 4.3 Tools, equipment and personal protective equipment 4.4 Appropriate workplace. 4.5 CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Performed Draining and replacing engine oil efficiently. 1.2 Cleaned and serviced engine oil filters and sumps. 1.3 Performed Disassembling, inspecting, and servicing engine oil pumps. 1.4 Checked, cleaned, and maintained engine oil coolers. 1.5 Used appropriate tools and following servicing procedures accurately.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module.

	3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.
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Unit of Competency: PERFORM COOLING SYSTEM SERVICING	Nominal Duration: 60 hrs.	Unit Code: SICIP-SBD-MEA-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform cooling system servicing. It specifically includes the tasks of carrying out cooling chamber maintenance, servicing cooling pump and servicing and maintaining water cooler.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out cooling chamber maintenance	1.1 <u>Cooling chamber</u> is inspected 1.2 Cooling chamber is cleaned and descaled 1.3 Sealing rings are checked and replaced if required
2. Service cooling pump	2.1 <u>Cooling pump</u> is checked and opened 2.2 Cooling pump is serviced 2.3 Cooling pump is fitted back
3. Service and maintain water cooler	3.1 <u>Water cooler</u> is checked and opened 3.2 Water cooler is serviced as per the standard operating procedure 3.3 Water cooler is cleaned

Range of Variables

Variable	Range May include but not limited to:
1 Cooling chamber	1.1 Cylinder Cooling Chamber 1.2 Piston Cooling Chamber 1.3 Exhaust Valve Cooling Chamber 1.4 Turbocharger Cooling Chamber
2 Cooling pump	2.1 Centrifugal Pump 2.2 Gear Pump 2.3 Reciprocating Pump 2.4 Jet Pump (Eductor Pump)
3 Water cooler	3.1 Plate type 3.2 Shell and tube type

Curricular Content Guide

1. Underpinning knowledge	1.1 Cooling System Components and Functions 1.2 Cooling Chamber Inspection and Maintenance 1.3 Cooling Pump Maintenance 1.4 Water Cooler Servicing and Cleaning 1.5 Standard operating procedures for cooling system servicing. 1.6 Workplace safety and environmental regulations related to cooling systems.
2. Underpinning Skills	2.1 Inspecting, cleaning, and descaling cooling chambers. 2.2 Checking, servicing, and reassembling cooling pumps. 2.3 Inspecting, cleaning, and maintaining water coolers. 2.4 Using appropriate tools and adhering to servicing procedures.
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness
4 Resource Implications	4.1. Marine operations site 4.2. Marine diesel engines and systems 4.3. Tools, equipment and personal protective equipment 4.4. Appropriate workplace. 4.5. CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Inspected, cleaned, and descaled cooling chambers. 1.2 Checked, serviced, and reassembled cooling pumps. 1.3 Inspected, cleaned, and maintained water coolers. 1.4 Used appropriate tools and adhering to servicing procedures
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: PERFORM MAINTENANCE OF INTAKE, EXHAUST SYSTEM	Nominal Duration: 40 hrs.	Unit Code: SICIP-SBD-MEA-05-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform maintenance of intake, exhaust system. It specifically includes the tasks of carrying out air cooler maintenance and carrying out exhaust manifold maintenance.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Carry out air cooler maintenance	1.1 Air cooler is checked and opened 1.2 Air cooler is serviced as per SOP 1.3 Air cooler is cleaned and fitted back
2. Carry out exhaust manifold maintenance	2.1 <u>Exhaust manifold</u> is checked and cleaned 2.2 Exhaust manifold expansion bellows is checked and 2.3 Exhaust manifold expansion bellows serviced as per SOP

Range of Variables

Variable	Range May include but not limited to:
1 Exhaust manifold	1.1 Water-Jacketed Exhaust Manifold 1.2 Dry-Type Expansion Chamber Manifold

Curricular Content Guide

1. Underpinning knowledge	1.1 Functions and types of air coolers and exhaust manifolds. 1.2 Proper cleaning and servicing techniques for intake and exhaust systems. 1.3 Common issues and failures in air coolers and exhaust manifolds. 1.4 Manufacturer guidelines and standard operating procedures (SOPs). 1.5 Workplace safety standards for handling exhaust system components.
2. Underpinning Skills	2.1 Inspecting, cleaning, and maintaining air coolers. 2.2 Checking and servicing exhaust manifolds and expansion bellows. 2.3 Following standard operating procedures (SOP) for maintenance tasks. 2.4 Reassembling components correctly after servicing.

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness
4. Resource Implications	4.1 Marine operations site 4.2 Marine diesel engines and systems 4.3 Tools, equipment and personal protective equipment 4.4 Appropriate workplace. 4.5 CBLM

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Inspected, cleaned, and maintained air coolers. 1.2 Checked and serviced exhaust manifolds and expansion bellows. 1.3 Followed standard operating procedures (SOP) for maintenance tasks. 1.4 Reassembled components correctly after servicing.
2. Methods of Assessment	Competency should be assessed by: 2.1 Written examination 2.2 Demonstration 2.3 Oral questioning 2.4 Workplace observation
3. Context of Assessment	3.1 Competency assessment must be done in a training center or in an actual or simulated work place after completion of the training module. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	Marine Engine Artificer
Number of Trainees:	25

Course-wise Training Space (Theoretical Classroom and Workshop):

- Classroom : 350 sft
- Workshop : 800-1000 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer with internet connection	1
2.	Projector and projector screen	1
3.	Marine Diesel engine	3
4.	Engine Cooler (Tube type)	1
5.	Engine Cooler (Plate type)	1
6.	Engine Crank shaft	1
7.	Different engine component set	2
8.	Centrifugal Pump	2
9.	Positive displacement pump	2
10.	Main bearing	1
11.	Crank pin bearing	1
12.	Battery (100 AH) with charger	1

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Marine Engine Artificer, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.